



Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For



Project Management Professional (PMP) Exam Prep

Course Code: C523

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version v2.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	20 July 2026	Initial release of the PMP Exam Prep courseware.	Dr. Alfred Ang
2.0	20 July 2026	Full-depth build: the deck now carries the complete ECO 2026 teaching content across all six topics (536 slides), with per-topic concept, process, formula, ITTO and exam-focus slides. Day 4 afternoon is given over to the capstone lab and the self-scored mock practice exam.	Dr. Alfred Ang

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How to Use This Guide

This Learner Guide accompanies the 4-day Project Management Professional (PMP) Exam Prep (course code C523), delivered by Dr. Alfred Ang. It follows the same six topics as the slide deck and the Lesson Plan, and the 24 hands-on labs referenced here are the same labs you complete in class.

Each topic chapter contains the teaching notes for that topic, the Examination Content Outline tasks it delivers, the labs that practise it, and a set of review questions. Work through the chapter after the session, then attempt the review questions without referring back.

The course is aligned to the PMI Examination Content Outline published for the July 2026 exam update. Where this guide differs from older PMP material, the 2026 outline governs.

Learning Outcomes

Ref	Learning Outcome
LO1	Scope medium-scale project requirements to drive timely completions.
LO2	Develop project plans based on realistic timelines and resource allocations.
LO3	Implement methodologies effectively to address project exigencies and derive solutions from identified root causes.
LO4	Analyze program risks and engage stakeholders through scheduled touchpoints to discuss potential issues.
LO5	Coordinate project deliverables against set objectives, costs, and timelines, and implement corrective actions.

Examination Content Outline — July 2026

Domain	Weighting	What it covers
Domain I. People	33%	How you lead, align and communicate with the humans on and around the project.
Domain II. Process	41%	How you plan, deliver, measure and close the work itself.
Domain III. Business Environment	26%	How the project connects to governance, compliance, risk and organisational change.

Exam format: 180 questions (170 scored, 10 unscored pretest) in 240 minutes. Two 10-minute breaks — the first after the case-study section, the second midway through the independent questions. About 40% predictive; the remaining 60% split between adaptive/agile and hybrid.

The Running Case Study

Every lab builds on the Contoso Training Portal Upgrade — a hybrid project to modernise a training provider's course registration and learner communications. It has a fixed launch date, a limited budget, a mandatory compliance review, and product

increments delivered in sprints. Artifacts you produce in early labs become inputs to later ones, so by the capstone you hold a complete project management plan.

Topic 1. Business Environment

Where projects come from, why they are funded, and the governance, compliance and change context they must survive in.

Examination Content Outline Coverage

Task	Task statement	Enablers
Business Environment T1	Define and establish project governance	Establish structure, rules, procedures, reporting, ethics and policies using OPAs; Define success metrics; Outline governance escalation paths and thresholds
Business Environment T2	Plan and manage project compliance	Confirm compliance requirements (security, health and safety, sustainability, regulatory); Classify compliance categories; Determine potential threats to compliance; Use methods to support compliance; Analyze the consequences of noncompliance; Determine the approach and actions to address compliance needs; Measure the extent to which the project is in compliance
Business Environment T7	Support organizational change	Assess organizational culture; Evaluate the impact of organizational change on the project and determine actions
Business Environment T8	Evaluate external business environment changes	Survey changes to the external business environment (regulations, technology, geopolitical, market); Assess and prioritize impact on project scope/backlog; Continually review the external business environment for impacts
Process T3	Help ensure value-based delivery	Identify value components with key stakeholders; Prioritize work based on value and stakeholder feedback; Assess opportunities to deliver value incrementally; Examine the business value throughout the project; Verify a measurement system is in place to track benefits; Evaluate delivery options to demonstrate value

What This Topic Covers

- Foundation and project fundamentals
- Strategic alignment and business value
- Development approaches: predictive, agile, hybrid
- Organisational culture and change management
- Project governance and escalation
- Compliance, sustainability and AI in projects

What a project is, and why that definition matters

A project is a temporary endeavour undertaken to create a unique product, service or result. Two words carry the weight. *Temporary* means it has a definite beginning and end — it is not the ongoing operation that follows. *Unique* means the deliverable differs in some material way from what came before; if you are producing the same thing repeatedly, that is operations, not a project. The exam tests this boundary constantly: maintaining a running payroll system is operations, replacing it is a project.

- Temporary — a defined start and finish, not perpetual
- Unique — the output differs materially from previous outputs
- Progressive elaboration — detail increases as the project advances
- Delivers business value, which is the reason it was funded at all

The triple constraint and why quality sits in the middle

Scope, time and cost form the classic triangle, with quality dependent on all three. Change any one and at least one other must move, or quality absorbs the damage. In practice this is the single most useful conversation tool a project manager has: when a stakeholder asks for more scope at a fixed date and budget, the triangle is what you draw. Modern practice extends this to six constraints — scope, schedule, cost, quality, resources and risk.

- Increase scope with fixed time and cost, and quality falls
- Compress schedule, and cost rises (crashing) or risk rises (fast tracking)
- Cut budget, and either scope or quality must give
- Never silently absorb a change — make the trade-off explicit and documented

Development approaches across the value delivery spectrum

Predictive (waterfall) approaches plan scope thoroughly up front and control change formally; they suit stable requirements and high-compliance environments. Adaptive (agile) approaches accept that requirements will emerge, and deliver in short increments so feedback arrives early. Hybrid blends them — for example, a fixed regulatory scope delivered predictively alongside a user-facing component delivered in sprints. The July 2026 exam splits roughly 40% predictive, 60% adaptive and hybrid, so you cannot pass on waterfall knowledge alone.

- Predictive — scope fixed early, change via formal change control
- Iterative — repeated cycles refine the same deliverable
- Incremental — successive slices each add usable functionality
- Agile — both iterative and incremental, with continuous feedback
- Hybrid — deliberately combines approaches per component

Enterprise environmental factors and organisational process assets

EEFs are conditions you do not control but must work within: market conditions, regulation, organisational culture, the infrastructure you are handed. OPAs are the assets your organisation gives you to reuse: templates, historical data, lessons learned,

procurement rules, defined processes. The exam distinction is simple but heavily tested — EEFs constrain you, OPAs help you.

- EEF internal — culture, structure, existing systems, staff capability
- EEF external — regulation, market conditions, standards, geopolitics
- OPA processes — policies, procedures, templates, defined workflows
- OPA knowledge — historical data, lessons learned, estimating databases

Governance, escalation and phase gates

Governance is the framework of authority: who decides what, within which thresholds, and what happens when a threshold is breached. A governance board or steering committee provides oversight; the project manager operates within delegated tolerances. When a problem exceeds your tolerance — a cost overrun beyond the agreed percentage, a risk above the agreed severity — you escalate. Phase gates are the scheduled decision points where the project is formally reviewed and authorised to continue, be redirected, or be stopped.

- Define escalation thresholds BEFORE you need them
- Escalating within an agreed threshold is competence, not failure
- Phase gates permit a go / no-go decision, including cancellation
- Governance defines success metrics, not just spending limits

Compliance, sustainability and AI — the 2026 additions

The July 2026 outline raised Business Environment from 8% to 26% of the exam and named sustainability explicitly across compliance, cost of quality and risk response. Compliance is no longer only legal and regulatory: environmental impact, workplace health and safety, ethical practice and data security all sit inside the compliance envelope. Artificial intelligence enters as both a tool (estimation, risk detection, reporting) and a governance concern (bias, explainability, data protection).

- Classify compliance categories, then assess threats to each
- Analyse the consequences of non-compliance, not just the requirement
- Measure the extent of compliance — it is a metric, not a binary
- AI used on a project needs its own governance and audit trail

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
01	PMP Exam Orientation and 35-PDU Study Plan	All domains	LO1	20 min	25 min	Personal study plan, Domain self-assessment
02	PESTLE and TECOP External Environment Scan	BE T8	LO1	20 min	25 min	PESTLE scan, TECOP risk themes
03	EEF/OPA Inventory with	BE T1	LO1	20 min	25 min	EEF/OPA inventory, Escalation

	Governance and Escalation Thresholds					threshold table
04	Compliance, Sustainability and AI Governance Register	BE T2	LO4	20 min	25 min	Compliance register, AI governance controls

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

1. Distinguish a project from operations, and give an example of each from your own workplace.
2. Your sponsor asks for additional scope while holding the date and budget fixed. Explain the trade-off using the triple constraint.
3. Classify each of the following as an EEF or an OPA: a lessons-learned database, a new data-protection regulation, the organisation's procurement policy, the existing IT infrastructure.
4. What is the difference between a governance escalation threshold and a phase gate?
5. Name three compliance categories beyond legal and regulatory that the 2026 outline expects you to consider.
6. How does the July 2026 exam update change the weighting of Business Environment, and why does that matter for your study plan?

Topic 2. Start the Project

Turning an idea into an authorised project: the business case, the charter, the stakeholders and the team.

Examination Content Outline Coverage

Task	Task statement	Enablers
People T1	Develop a common vision	Help ensure a shared vision with key stakeholders; Promote the shared vision; Keep the vision current; Break down situations to identify the root cause of a misunderstanding of the vision
People T4	Engage stakeholders	Identify stakeholders; Analyze stakeholders; Analyze and tailor communication to stakeholder needs; Execute the stakeholder engagement plan; Optimize alignment among stakeholder needs, expectations, and project objectives; Build trust and influence stakeholders to accomplish project objectives
People T5	Align stakeholder expectations	Categorize stakeholders; Identify stakeholder expectations; Facilitate discussions to align expectations; Organize and act on mentoring opportunities
People T6	Manage stakeholder expectations	Identify internal and external customer expectations; Align and maintain outcomes to internal and external customer expectations; Monitor internal and external customer satisfaction/expectations and respond as needed
Process T1	Develop an integrated project management plan and plan delivery	Assess project needs, complexity, and magnitude; Recommend a development approach (predictive, adaptive/agile, or hybrid); Determine critical information requirements (e.g., sustainability); Recommend a project execution strategy; Create an integrated project management plan; Estimate work effort and resource requirements; Assess consolidated plans for dependencies, gaps, and continued business value; Maintain the integrated project management plan; Collect and analyze data to make informed project decisions

What This Topic Covers

- Project integration and the charter

- Business case, cost-benefit and benefits management
- Identify, analyse and prioritise stakeholders
- Align and manage stakeholder expectations
- Form the team and build shared understanding
- Determine the project approach

The business case and why projects get funded

Projects exist to deliver benefit. The business case is the documented economic feasibility study that justifies the investment: the problem or opportunity, the options considered, the recommended option, and the financial and non-financial returns. It is prepared before the charter and is the reference point you return to whenever someone proposes a change that erodes the value the project was funded to deliver.

- Cost-benefit analysis compares the cost of the work to the benefit gained
- Payback period — how long until the investment is recovered; shorter is better
- Net present value (NPV) — future cash flows discounted to today; positive is good
- Internal rate of return (IRR) and benefit-cost ratio (BCR) — higher is better
- Opportunity cost — the value of the option you did NOT choose

The project charter

The charter authorises the project and, critically, authorises the project manager to apply organisational resources to it. It is issued by a sponsor external to the project with the authority to fund it. Without a charter you have no mandate. It stays deliberately high level — names, objectives, high-level requirements, summary milestones and budget, key risks, and the approval criteria.

- Authorises the project's existence and the PM's authority
- Signed by the sponsor, not written by the sponsor alone
- Contains measurable objectives and success criteria
- Records assumptions, constraints and high-level risk
- Is not re-baselined — it is the founding document

Identifying and analysing stakeholders

A stakeholder is anyone who affects, is affected by, or believes they are affected by the project. That last clause matters: perceived impact generates real behaviour. Identification is continuous, not a one-off task at initiation. The stakeholder register records who they are, their interest, influence, expectations and your engagement strategy.

- Power/interest grid — the most commonly examined classification
- Salience model — power, legitimacy and urgency combined
- Directions of influence — upward, downward, sideward, outward
- Engagement levels — unaware, resistant, neutral, supportive, leading
- The SEAM records Current (C) versus Desired (D) engagement per stakeholder

Aligning and managing expectations

The 2026 outline separates engaging stakeholders (People T4) from aligning their expectations (T5) and managing those expectations over time (T6). Alignment is the facilitation work of getting parties with different definitions of success to agree one definition. Management is the ongoing monitoring of satisfaction and responding when it drifts.

- Identify expectations explicitly — unstated expectations become disputes
- Facilitate discussion where expectations conflict; do not arbitrate silently
- Monitor internal and external customer satisfaction continuously
- Build trust deliberately; influence follows trust, not authority

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
05	Business Case and Cost-Benefit Analysis	Process T3	LO1	25 min	25 min	Business case, NPV/ROI/payback model
06	Project Charter	Process T1	LO1	20 min	25 min	Approved project charter
07	Stakeholder Register, Power/Interest Grid and Salience Model	People T4, T5	LO4	25 min	25 min	Stakeholder register, Power/interest grid, Salience classification
08	Team Charter, Ground Rules and Shared Vision	People T1, T2	LO3	30 min	15 min	Team charter, Ground rules, Vision statement

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

7. What does the project charter authorise that no other document does?
8. A project has an initial investment of \$200,000 and returns \$60,000 per year. Calculate the payback period and state one limitation of using it alone.
9. Compare the power/interest grid with the salience model. When would you prefer each?
10. A stakeholder is currently resistant but needs to be supportive. How would you record and act on that in a SEAM?
11. Explain the difference between engaging stakeholders and aligning their expectations, per ECO People T4 and T5.
12. Why is 'anyone who believes they are affected' included in the definition of a stakeholder?

Topic 3. Plan the Project

Building the integrated plan — scope, schedule, cost, quality, resources, communications, risk and procurement.

Examination Content Outline Coverage

Task	Task statement	Enablers
Process T1	Develop an integrated project management plan and plan delivery	Assess project needs, complexity, and magnitude; Recommend a development approach (predictive, adaptive/agile, or hybrid); Determine critical information requirements (e.g., sustainability); Recommend a project execution strategy; Create an integrated project management plan; Estimate work effort and resource requirements; Assess consolidated plans for dependencies, gaps, and continued business value; Maintain the integrated project management plan; Collect and analyze data to make informed project decisions
Process T2	Develop and manage project scope	Define scope; Obtain stakeholder agreement on project scope; Break down scope
Process T4	Plan and manage resources	Define and plan resources based on requirements; Manage and optimize resource needs and availability
Process T5	Plan and manage procurement	Plan procurement; Execute a procurement management plan; Select preferred contract types; Evaluate vendor performance; Verify objectives of the procurement agreement are met; Participate in agreement negotiations; Determine a negotiation strategy; Manage suppliers and contracts; Plan and manage the procurement strategy; Develop a delivery solution
Process T6	Plan and manage finance	Analyze project financial needs; Quantify risk and contingency financial allocations; Plan spend tracking throughout the project life cycle; Plan financial reporting; Anticipate future finance challenges; Monitor financial variations and work with the governance process; Manage financial reserves
Process T7	Plan and optimize quality of products/deliverables	Gather quality requirements for project deliverables; Plan quality processes and tools; Execute a

		quality management plan; Help ensure regulatory compliance; Manage cost of quality (CoQ) and sustainability; Conduct ongoing quality reviews; Implement continuous improvement
Process T8	Plan and manage schedule	Prepare a schedule based on the selected development approach; Coordinate with other projects and operations; Estimate project tasks (milestones, dependencies, story points); Utilize benchmarks and historical data; Create a project schedule; Baseline a project schedule; Execute a schedule management plan; Analyze schedule variation
Business Environment T5	Plan and manage risk	Identify risks; Analyze risks; Monitor and control risks; Develop a risk management plan; Maintain a risk register (e.g., poor IT security); Execute a risk management plan (risk response for security and sustainability risks); Communicate the status of a risk impact on the project
People T8	Plan and manage communication	Define a communication strategy; Promote transparency and collaboration; Establish a feedback loop; Understand reporting requirements; Create reports aligned with sponsors and stakeholder expectations; Support reporting and governance processes

What This Topic Covers

- The integrated project management plan
- Scope: requirements, WBS and baselines
- Schedule: activities, dependencies, critical path
- Cost: estimating, budget and reserves
- Quality: planning and problem-solving tools
- Resources, communications and procurement
- Risk: identification, analysis and response

The integrated project management plan

The project management plan is not one document but the integration of all subsidiary plans — scope, schedule, cost, quality, resource, communications, risk, procurement and stakeholder engagement — plus the three baselines (scope, schedule, cost). Integration is the point: the plans must be internally consistent, and a change to one usually forces a change to others.

- Subsidiary management plans define HOW each knowledge area will be run

- Baselines are the approved versions against which performance is measured
- Changes to a baseline require an approved change request — always
- Rolling wave planning elaborates near-term work in detail, later work coarsely

Scope: from requirements to work packages

Scope work moves from elicitation (what do stakeholders need?) through definition (what will we deliver?) to decomposition (what work produces it?). The Requirements Traceability Matrix links each requirement forward to the deliverable, test and acceptance that satisfies it. The WBS decomposes the deliverables — not the activities — down to work packages, and obeys the 100% rule: the WBS contains all the work and only the work.

- Scope creep — uncontrolled additions without change control
- Gold plating — the team adding unrequested extras; never acceptable
- The WBS decomposes DELIVERABLES; the activity list decomposes the work
- The WBS dictionary carries the detail for each work package
- Scope baseline = scope statement + WBS + WBS dictionary

Schedule: dependencies, estimating and the critical path

Activities are sequenced by dependency type (mandatory, discretionary, external, internal) and relationship (finish-to-start being the most common). Durations are estimated by analogous, parametric, three-point or bottom-up methods. The critical path is the longest path through the network and therefore the shortest possible duration; activities on it have zero float. Compressing it means crashing (adding cost) or fast tracking (adding risk).

- Three-point (PERT): $(O + 4M + P) / 6$, with $\sigma = (P - O) / 6$
- Total float — delay available before the project end date slips
- Free float — delay available before the NEXT activity is affected
- Critical path activities have zero total float by definition
- Crashing adds cost; fast tracking adds risk and potential rework

Cost: estimating, budgeting and reserves

Cost estimates roll up from work packages into the cost baseline, to which contingency reserve is added for identified risks. Management reserve sits above the cost baseline for unknown-unknowns and requires management approval to access. The distinction is examined frequently: contingency is inside the baseline and under your control, management reserve is outside it and is not.

- Cost baseline = work package estimates + contingency reserve
- Budget at completion (BAC) = cost baseline + management reserve
- Contingency reserve — for identified (known) risks; PM controls it
- Management reserve — for unidentified risks; management controls it

Quality: planning it in, not inspecting it in

Quality is conformance to requirements and fitness for use. It is cheaper to prevent than to inspect, and far cheaper to fix early than late — the cost of change curve rises steeply as the project progresses. The cost of quality splits into conformance costs (prevention, appraisal) and non-conformance costs (internal failure, external failure). External failure — the customer finds the defect — is the most expensive outcome of all.

- Prevention — training, process design, prototyping; the cheapest lever
- Appraisal — inspection, testing, audits; finds defects before the customer
- Internal failure — rework caught in-house
- External failure — the customer finds it; cost includes reputation
- Root cause tools: 5 Whys, fishbone, Pareto, control charts

Risk: identification through response

Risk is an uncertain event that, if it occurs, has a positive or negative effect on objectives. Risks are future and uncertain; issues are present and certain. Qualitative analysis prioritises risks by probability and impact; quantitative analysis models their aggregate effect in money or time. Response strategies mirror each other for threats and opportunities.

- Threats: escalate, avoid, transfer, mitigate, accept
- Opportunities: escalate, exploit, share, enhance, accept
- $EMV = \text{probability} \times \text{impact}$, summed across outcomes
- Secondary risk arises FROM your response; residual risk remains after it
- A risk that has occurred is no longer a risk — it is an issue

Procurement and contract types

Contract type determines who carries cost risk. Fixed-price contracts place risk on the seller and suit well-defined scope. Cost-reimbursable contracts place risk on the buyer and suit uncertain scope. Time and materials sits between the two and suits small or urgent engagements where scope is not yet clear.

- Firm fixed price (FFP) — seller carries the cost risk entirely
- Fixed price incentive fee (FPIF) — shared risk with a performance incentive
- Cost plus fixed fee (CPFF) — buyer carries cost risk; fee is fixed
- Cost plus incentive/award fee (CPIF/CPAF) — buyer risk with incentives
- Time and materials (T&M) — hybrid; cap it or it becomes open-ended

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
09	Requirements Elicitation and Traceability Matrix	Process T2	LO1	30 min	20 min	Requirements list, RTM
10	MoSCoW and Kano Prioritisation	Process T3	LO1	30 min	20 min	Prioritised backlog, Story point

	with Product Backlog					estimates
11	Work Breakdown Structure and WBS Dictionary	Process T2	LO1	30 min	20 min	WBS, WBS dictionary, Scope baseline
12	Network Diagram, PERT Estimating and Critical Path	Process T8	LO2	35 min	25 min	Network diagram, Critical path, Float table
13	Cost Estimating, Budget and Reserve Analysis	Process T6	LO2	25 min	25 min	Cost estimates, Cost baseline, Reserve plan
14	Risk Register, Probability/Impact Matrix and EMV Decision Tree	BE T5	LO4	30 min	30 min	Risk register, P/I matrix, EMV decision tree

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

13. State the 100% rule and explain what it prohibits.
14. Given O=4, M=6, P=14 days, calculate the PERT estimate and the standard deviation.
15. Distinguish total float from free float, and state the float of an activity on the critical path.
16. A risk has a 30% probability and a \$80,000 impact. Calculate the EMV and explain how you would use it.
17. Contrast contingency reserve with management reserve, including who controls each.
18. Which contract type would you choose for a well-defined deliverable with a fixed specification, and who carries the cost risk?
19. Distinguish scope creep from gold plating, and state why gold plating is never acceptable.
20. List the five threat response strategies and their opportunity mirrors.

Topic 4. Lead the Project Team

The People domain in practice — leadership style, motivation, conflict, coaching and communication.

Examination Content Outline Coverage

Task	Task statement	Enablers
People T1	Develop a common vision	Help ensure a shared vision with key stakeholders; Promote the shared vision; Keep the vision current; Break down situations to identify the root cause of a misunderstanding of the vision
People T2	Manage conflicts	Identify conflict sources; Analyze the context for the conflict; Implement an agreed-on resolution strategy; Communicate conflict management principles with the team and external stakeholders; Establish an environment that fosters adherence to common ground rules; Manage and rectify ground rule violations
People T3	Lead the project team	Establish expectations at the team level; Empower the team; Solve problems; Represent the voice of the team; Support the team's varied experiences, skills, and perspectives; Determine an appropriate leadership style; Establish clear roles and responsibilities within the team
People T7	Help ensure knowledge transfer	Identify knowledge critical to the project; Gather knowledge; Foster an environment for knowledge transfer
People T8	Plan and manage communication	Define a communication strategy; Promote transparency and collaboration; Establish a feedback loop; Understand reporting requirements; Create reports aligned with sponsors and stakeholder expectations; Support reporting and governance processes
Process T1	Develop an integrated project management plan and plan delivery	Assess project needs, complexity, and magnitude; Recommend a development approach (predictive, adaptive/agile, or hybrid); Determine critical information requirements (e.g., sustainability); Recommend a project execution strategy; Create an integrated project management plan; Estimate work effort and resource requirements; Assess consolidated plans for

		dependencies, gaps, and continued business value; Maintain the integrated project management plan; Collect and analyze data to make informed project decisions
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What This Topic Covers

- Direct and manage project work
- Craft your leadership skills
- Build a collaborative team environment
- Empower the team and support performance
- Training, coaching and mentoring
- Manage conflict and negotiate

Leadership style and the servant leader

There is no single correct leadership style; the ECO asks you to *determine an appropriate* style for the situation, team maturity and organisational context. Servant leadership — the dominant model in adaptive environments — inverts the usual hierarchy: the leader's job is to remove impediments, grow people and make it safe to raise problems, so the team can do the work.

- Servant leadership — serve the team so the team can deliver
- Transformational — inspire through vision and individual attention
- Transactional — exchange reward for defined performance
- Laissez-faire — hands off; only works with a mature, self-organising team
- Situational — deliberately vary the style by person and circumstance

Team development and motivation

Tuckman's model — forming, storming, norming, performing, adjourning — describes how teams mature, and each stage calls for different leadership. Storming is normal and necessary, not a failure. Motivation theory tells you what to reach for: Herzberg separates hygiene factors (whose absence demotivates) from motivators (whose presence motivates), which is why a pay rise fixes dissatisfaction but does not create engagement.

- Forming — provide direction and clarify purpose
- Storming — facilitate conflict; do not suppress it
- Norming — reinforce the working agreements the team has built
- Performing — delegate and remove blockers; stay out of the way
- Adjourning — recognise contribution and capture lessons learned

Conflict management — the most examined People topic

Conflict is inevitable and, handled well, productive. The five modes trade off assertiveness against cooperativeness. Collaborating (problem solving) produces a genuine win-win and is the preferred answer in most exam situations, because it addresses the underlying need rather than splitting the difference. Forcing and avoiding

are appropriate only in narrow circumstances — emergencies, or trivia not worth the cost of resolution.

- Collaborate / problem solve — win-win; the usual best answer
- Compromise / reconcile — lose-lose; both give something up
- Smooth / accommodate — lose-win; preserves the relationship
- Force / direct — win-lose; fast, but damages trust
- Withdraw / avoid — defers; appropriate only for trivia or cooling off

Communication

Communication consumes most of a project manager's time. The channels formula, $N(N-1)/2$, shows why: adding people to a team increases the communication burden quadratically. A communications management plan defines who needs what information, in what format, how often, and through which method — push, pull or interactive.

- Channels = $N(N - 1) / 2$ — ten people means forty-five channels
- Push — sent to recipients (email, reports); no confirmation of understanding
- Pull — recipients retrieve it (portals, dashboards); suits large audiences
- Interactive — real-time exchange; the only mode that confirms understanding
- Tailor the message to the stakeholder — this is an explicit ECO enabler

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
15	RACI Responsibility Assignment Matrix	People T3, Process T4	LO2	20 min	25 min	RACI matrix, Resource plan
16	Tuckman Diagnosis and Motivation Theory Application	People T3	LO3	20 min	25 min	Team stage diagnosis, Motivation action plan
17	Conflict Resolution Role-Play across the Five Modes	People T2	LO3	20 min	30 min	Conflict analysis, Resolution strategy
18	Communication Plan, Channels Formula and Status Report	People T8	LO4	20 min	30 min	Communications management plan, Status report

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

21. Describe the leadership action appropriate at each stage of Tuckman's model.
22. Explain why a pay rise may remove dissatisfaction without creating motivation, using Herzberg's theory.

23. A designer and a developer disagree on a screen layout. Which conflict mode would you use first, and why?
24. Calculate the number of communication channels for a team of 12, and state what the result implies for your communications plan.
25. Distinguish push, pull and interactive communication, and state which confirms understanding.
26. What does 'represent the voice of the team' require of a project manager in practice?

Topic 5. Monitor and Control the Project

Measuring truth against plan: earned value, quality control, change control, issues and impediments.

Examination Content Outline Coverage

Task	Task statement	Enablers
Process T9	Evaluate project status	Develop project metrics, analysis, and reconciliation; Identify and tailor needed artifacts; Help ensure artifacts are created, reviewed, updated, and documented; Help ensure accessibility of artifacts; Assess current progress; Measure, analyze, and update project metrics; Communicate project status; Continually assess the effectiveness of artifact management
Business Environment T3	Manage and control changes	Execute the change control process; Communicate the status of proposed changes; Implement approved changes to the project; Update project documentation to reflect changes
Business Environment T4	Remove impediments and manage issues	Evaluate the impact of impediments; Prioritize and highlight impediments; Determine and apply an intervention strategy to remove/minimize impediments; Reassess continually to ensure blockers are being addressed; Recognize when a risk becomes an issue; Collaborate with relevant stakeholders on an approach to resolve issues
Business Environment T5	Plan and manage risk	Identify risks; Analyze risks; Monitor and control risks; Develop a risk management plan; Maintain a risk register (e.g., poor IT security); Execute a risk management plan (risk response for security and sustainability risks); Communicate the status of a risk impact on the project
Business Environment T6	Continuous improvement	Utilize lessons learned; Help ensure continuous improvement processes are updated; Update organizational process assets (OPAs)
Process T7	Plan and optimize quality of products/deliverables	Gather quality requirements for project deliverables; Plan quality processes and tools; Execute a quality management plan; Help ensure regulatory compliance; Manage cost of quality (CoQ) and sustainability; Conduct

		ongoing quality reviews; Implement continuous improvement
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What This Topic Covers

- Monitor and control project work
- Evaluate project progress and earned value
- Control quality with statistical tools
- Manage changes and integrated change control
- Manage issues, impediments and risk
- Implement ongoing improvements

Earned value management

EVM integrates scope, schedule and cost into a single set of measures, and it is the most calculation-heavy topic on the exam. Three inputs drive everything: planned value (what you said you would have done by now), earned value (what you have actually completed, valued at budget) and actual cost (what you spent doing it). Everything else derives from those three.

- $CV = EV - AC$ · $SV = EV - PV$ (positive is good in both cases)
- $CPI = EV / AC$ · $SPI = EV / PV$ (above 1.0 is good in both cases)
- $EAC = BAC / CPI$ when current variances are expected to continue
- $ETC = EAC - AC$ · $VAC = BAC - EAC$
- $TCPI = (BAC - EV) / (BAC - AC)$ — the performance needed to finish on budget

Integrated change control

Every change request follows the same path regardless of who raised it: record it, assess its impact across scope, schedule, cost, quality and risk, take it to the change control board for a decision, then either implement and update the baselines and documents, or record the rejection. The exam's most common trap is the option that implements a change without assessing impact or obtaining approval — it is always wrong.

- Never implement an unapproved change, however senior the requester
- Assess impact across ALL constraints before recommending a decision
- Corrective action realigns performance with the plan
- Preventive action reduces the probability of future variance
- Update the baselines and affected documents after approval

Controlling quality with statistical tools

Control charts distinguish variation that is inherent in the process (common cause) from variation that signals something has changed (assignable cause). A point outside the control limits is out of control. So is the rule of seven: seven consecutive points on one side of the mean indicates a non-random pattern even when every point sits inside the limits.

- Control limits (UCL/LCL) are calculated from the process, usually $\pm 3\sigma$
- Specification limits come from the customer and are a different thing
- Rule of seven — seven consecutive points on one side signals assignable cause
- Pareto — the 80/20 rule; fix the vital few categories first
- In control does not mean acceptable; it means predictable

Issues, impediments and risk that has arrived

The 2026 outline gives impediment removal its own task. The distinction matters: a risk is uncertain and future; when it occurs it becomes an issue and moves from the risk register to the issue log. Impediments and blockers are anything stopping the team from progressing, and clearing them is explicitly the project manager's job.

- Recognise the moment a risk becomes an issue and re-classify it
- Evaluate impact, then prioritise — not every impediment is urgent
- Apply an intervention strategy, then reassess that it actually worked
- Escalate beyond your threshold rather than absorbing the impact silently

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
19	Kanban Board with WIP Limits and Lead/Cycle Time	Process T9	LO5	20 min	25 min	Kanban board, Cycle time measurements
20	Earned Value Management and Schedule Compression	Process T9	LO5	25 min	35 min	EVM calculation sheet, Forecast (EAC/ETC), Compression decision
21	Root Cause Analysis with 5 Whys, Fishbone and Pareto	BE T4	LO3	25 min	35 min	5 Whys chain, Fishbone diagram, Pareto chart
22	SPC Control Chart and Statistical Process Analysis	Process T7	LO5	30 min	30 min	Control chart, Stability verdict, Capability summary

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

27. A project has BAC \$500,000. At week 10: PV \$200,000, EV \$180,000, AC \$210,000. Calculate CV, SV, CPI, SPI and EAC, and state the project's health in one sentence.
28. A senior stakeholder instructs you to implement a change immediately. State your correct sequence of actions.

29. Distinguish common cause from assignable cause variation, and explain the rule of seven.
30. When does a risk become an issue, and what changes in your documentation at that moment?
31. Compare crashing with fast tracking, including the cost and risk consequences of each.
32. Explain how a Pareto chart directs your quality improvement effort.

Topic 6. Close the Project

Landing the project: acceptance, transition, benefits realisation and the knowledge that outlives the team.

Examination Content Outline Coverage

Task	Task statement	Enablers
Process T10	Manage project closure	Obtain project stakeholder approval of project completion; Determine criteria to successfully close the project or phase; Validate readiness for transition (e.g., to operations or next phase); Conclude closure activities (lessons learned, retrospectives, procurement, financials, resources)
Business Environment T6	Continuous improvement	Utilize lessons learned; Help ensure continuous improvement processes are updated; Update organizational process assets (OPAs)
People T7	Help ensure knowledge transfer	Identify knowledge critical to the project; Gather knowledge; Foster an environment for knowledge transfer

What This Topic Covers

- Project and phase closure
- Benefits realisation and value confirmation
- Knowledge transfer and lessons learned
- PMP exam strategy and readiness

Closing the project or phase

Closure is a formal process, not simply stopping work. Deliverables are accepted against the acceptance criteria agreed at the start; the product is transitioned to the receiving organisation with the knowledge needed to run it; procurements are closed and contracts archived; resources are released; and lessons learned are transferred into the organisational repository so the next project inherits them. Projects terminated early are closed through exactly the same process.

- Obtain formal stakeholder approval of completion — not informal agreement
- Validate readiness for transition before handing over
- Archive contracts, financial records and project documentation
- Release resources deliberately, with recognition
- Update OPAs so the organisation learns; this is the durable output

Benefits realisation

The project ends before the benefits arrive. The benefits management plan names the benefits owner, the metrics and the measurement timeframe, so that value is confirmed after handover rather than assumed at closure. The 2026 outline's emphasis on value-

based delivery makes this explicit: verify a measurement system is in place to track benefits.

- Benefits are usually realised after the project closes
- A named benefits owner carries accountability past project end
- Metrics and baselines must exist before you can claim value
- Examine business value throughout — not only at the gate reviews

Preparing for the examination

The exam is 180 questions in 240 minutes: about eighty seconds each. Ten questions are unscored pretest items, indistinguishable from the rest. Two ten-minute breaks are offered, the first after the case-study section — and once you start a break you cannot return to the previous section, so review before you break, not after.

- 180 questions (170 scored, 10 pretest) in 240 minutes
- People 33% · Process 41% · Business Environment 26%
- About 40% predictive, 60% adaptive and hybrid
- Six question types, including the new graphic-based items
- Read for what the question ASKS — 'first', 'best', 'next' change the answer

Hands-on Labs for This Topic

Lab	Title	ECO	LO	In class	Self-study	Produces
23	Project Closure, Benefits Realisation and Lessons Learned	Process T10, BE T6	LO5	105 min	0 min	Closure checklist, Final report, Lessons learned register
24	Capstone: Consolidated Project Plan and Mock Exam	All domains	LO1-LO5	175 min	0 min	Consolidated project management plan, Mock exam score analysis by domain

Full step-by-step instructions for each lab are in the labs/ folder of your course materials. You complete the core steps in class; the extension steps are self-study that finishes the artifact.

Review Questions

33. List the activities required to formally close a project or phase.
34. Why are benefits usually realised after the project has closed, and who is accountable for them?
35. What is the difference between the lessons learned register and the lessons learned repository?
36. A project is terminated early. Which closure activities still apply?
37. State the exam's question count, scored count, duration and break structure.
38. Explain the significance of the words 'first', 'best' and 'next' in a situational exam question.

Complete Lab Index

24 labs totalling 1385 minutes (23.1 hours) of hands-on practice: 820 minutes timetabled in class across the four days, and 565 minutes of extension work you complete as self-study.

Lab	Title	Day	Topic	ECO Task	LO	In class	Self-study
01	PMP Exam Orientation and 35-PDU Study Plan	Day 1	T1	All domains	LO1	20	25
02	PESTLE and TECOP External Environment Scan	Day 1	T1	BE T8	LO1	20	25
03	EEF/OPA Inventory with Governance and Escalation Thresholds	Day 1	T1	BE T1	LO1	20	25
04	Compliance, Sustainability and AI Governance Register	Day 1	T1	BE T2	LO4	20	25
05	Business Case and Cost-Benefit Analysis	Day 1	T2	Process T3	LO1	25	25
06	Project Charter	Day 1	T2	Process T1	LO1	20	25
07	Stakeholder Register, Power/Interest Grid and Salience Model	Day 1	T2	People T4, T5	LO4	25	25
08	Team Charter, Ground Rules and Shared Vision	Day 1	T2	People T1, T2	LO3	30	15
09	Requirements Elicitation and	Day 2	T3	Process T2	LO1	30	20

	Traceability Matrix						
10	MoSCoW and Kano Prioritisation with Product Backlog	Day 2	T3	Process T3	LO1	30	20
11	Work Breakdown Structure and WBS Dictionary	Day 2	T3	Process T2	LO1	30	20
12	Network Diagram, PERT Estimating and Critical Path	Day 2	T3	Process T8	LO2	35	25
13	Cost Estimating, Budget and Reserve Analysis	Day 2	T3	Process T6	LO2	25	25
14	Risk Register, Probability /Impact Matrix and EMV Decision Tree	Day 2	T3	BE T5	LO4	30	30
15	RACI Responsibility Assignment Matrix	Day 3	T4	People T3, Process T4	LO2	20	25
16	Tuckman Diagnosis and Motivation Theory Application	Day 3	T4	People T3	LO3	20	25
17	Conflict Resolution Role-Play across the Five Modes	Day 3	T4	People T2	LO3	20	30
18	Communication Plan, Channels Formula and Status Report	Day 3	T4	People T8	LO4	20	30
19	Kanban Board with WIP Limits	Day 3	T5	Process T9	LO5	20	25

	and Lead/Cycle Time						
20	Earned Value Management and Schedule Compression	Day 3	T5	Process T9	LO5	25	35
21	Root Cause Analysis with 5 Whys, Fishbone and Pareto	Day 3	T5	BE T4	LO3	25	35
22	SPC Control Chart and Statistical Process Analysis	Day 3	T5	Process T7	LO5	30	30
23	Project Closure, Benefits Realisation and Lessons Learned	Day 4	T6	Process T10, BE T6	LO5	105	0
24	Capstone: Consolidated Project Plan and Mock Exam	Day 4	T6	All domains	LO1-LO5	175	0

Tools Used in the Labs

Tool	Purpose	URL
5 Whys	Drill from a symptom to its root cause by asking 'why' five times.	https://alfredang.github.io/5whys/
Fishbone Diagram	Group candidate causes into categories around a problem statement.	https://alfredang.github.io/fishbone/
Pareto Chart	Rank defect categories to find the vital few driving most of the pain.	https://alfredang.github.io/paretochart/
System Thinking	Map reinforcing and balancing loops behind a recurring problem.	https://alfredang.github.io/systemloop/
SPC / Control Chart	Plot a process over time against control limits to separate signal from noise.	https://alfredang.github.io/novaspc/
Statistics	Descriptive statistics, distributions and hypothesis testing on project data.	https://alfredang.github.io/novastats/
RACI Matrix	Assign Responsible, Accountable, Consulted and Informed per work package.	https://alfredang.github.io/raci/
Kanban Board	Visualise work in progress, limit	https://alfredang.github.io/

	WIP and expose bottlenecks.	kanban/
Scrum Simulator	Run sprint planning, review and retrospective mechanics.	https://alfredang.github.io/scrum/
Mind Mapping	Expand a central idea into branches for requirements discovery.	https://alfredang.github.io/mindmapping/
Design Thinking	Empathise, define, ideate, prototype and test a solution.	https://alfredang.github.io/designthinking/
Pivot Analysis	Slice project data by dimension to expose trends and outliers.	https://alfredang.github.io/novapivot/

Preparing for the PMP Examination

Exam Structure

Element	Detail
Questions	180 (170 scored + 10 pretest)
Duration	240 minutes
Breaks	Two 10-minute breaks — the first after the case-study section, the second midway through the independent questions.
Approach mix	About 40% predictive; the remaining 60% split between adaptive/agile and hybrid.
Domains	People 33% · Process 41% · Business Environment 26%

Question Types

Type	What it asks of you
Multiple choice	One best answer from four options — the bulk of the exam.
Multiple response	More than one correct answer; the item tells you how many.
Matching	Drag items from one column to match a second column.
Enhanced matching	Matching that includes an image or diagram for context.
Graphic-based (NEW)	Read a chart, graph, diagram or image, then answer from it.
Case study	A scenario with supporting visuals, then a series of linked questions.

Formula Reference

Formula	Expression	Reading
Cost variance	$CV = EV - AC$	Positive is under budget
Schedule variance	$SV = EV - PV$	Positive is ahead of schedule
Cost performance index	$CPI = EV / AC$	Above 1.0 is under budget
Schedule performance index	$SPI = EV / PV$	Above 1.0 is ahead
Estimate at completion	$EAC = BAC / CPI$	When current variance is expected to continue
Estimate to complete	$ETC = EAC - AC$	Remaining expected cost
Variance at completion	$VAC = BAC - EAC$	Forecast budget variance
To-complete performance index	$TCPI = (BAC - EV) / (BAC - AC)$	Efficiency needed to finish on budget
Three-point (PERT)	$(O + 4M + P) / 6$	Weighted average estimate
PERT standard deviation	$(P - O) / 6$	Estimate uncertainty
Expected monetary value	$EMV = \text{probability} \times \text{impact}$	Summed across outcomes
Communication channels	$N(N - 1) / 2$	Grows quadratically

Study Plan After the Course

39. Week 1-2: Re-read this guide topic by topic and redo the labs on a project of your own.
40. Week 3-4: Drill practice questions by domain, weighting your time Process 41%, People 33%, Business Environment 26%.
41. Week 5-6: Practise the graphic-based and case-study question types specifically — they behave differently from plain multiple choice.

42. Week 7-8: Sit at least three full 180-question mock exams under timed conditions, reviewing every wrong answer to its underlying concept.
43. Submit your PMI application using this course for the 35 contact hours.